SMART CROP ADVISORY SYSTEM

PROJECT REPORT

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of

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in

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Department of Computer Science and Engineering

College of Engineering Trivandrum

Kerala

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DECLARATION

We undersigned hereby declare that the project report **SMART CROP ADVISORY SYSTEM**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Prof. Anoopa S.**. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Thiruvananthapuram

Date: April, 2026

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CERTIFICATE

This is to certify that the report entitled **"SMART CROP ADVISORY SYSTEM",** submitted by **Akshara .S** (TVE23CS019), **Maria Binu Simon** (TVE23CS090), **Nida Farheen** (TVE23CS104), **Nikhil Prakash J.** (LTVE23CS151) to the **APJ Abdul Kalam Technological University** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Agriculture today requires informed decision-making that considers soil conditions, climatic factors, financial constraints, and market trends. However, many farmers still rely on traditional practices and limited information while selecting crops and planning cultivation, which can lead to reduced productivity and economic loss.The Smart Crop Advisory System aims to address these challenges by providing an integrated, data-driven advisory platform for farmers. The system recommends suitable crops based on soil type, climate, location, and budget using machine learning techniques. It also provides real-time weather-based alerts through third-party weather APIs and market advisory to support better selling decisions. A dynamic crop growth planner is included to guide farmers through different crop growth stages.The proposed solution is implemented as a web-based application using Django for backend processing, machine learning models for crop recommendation, and PostgreSQL for data storage.At the current stage, requirement analysis, literature survey, and system design have been completed, and implementation has been initiated with development of the first module of the system.

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ABBREVIATIONS

AI Artificial Intelligence

ML Machine Learning

NOTATION

K Potassium

N Nitrogen

P Phosphorus

pH Measure of acidity or alkalinity

Chapter 1

Introduction

In recent years, the use of data-driven technologies has significantly improved decision-making across various domains, including farming. One of the major challenges faced by farmers is selecting the most suitable crop and managing it effectively under varying soil conditions, weather patterns, and market fluctuations The proposed Smart Crop Recommendation System addresses this problem by providing intelligent crop suggestions based on key parameters such as soil nutrients (N, P, K), pH, temperature, humidity, and rainfall. The system uses Machine Learning (ML) techniques, specifically the Random Forest algorithm, to analyze these factors and recommend the most appropriate crops. In addition to crop prediction, the system offers an advisory feature that provides stage-wise guidance throughout the crop lifecycle. It also includes features such as yield prediction and market insights to support better planning and decision-making. Overall, this project demonstrates how technology can be used to make farming more efficient, informed, and data-driven.

Chapter 2

Existing Methods

Traditional farming methods mainly rely on farmers' experience, intuition, and knowledge passed down through generations. Decisions such as crop selection, irrigation practices, and harvesting time are often made based on past practices rather than scientific analysis. While these methods have been effective for many years, they lack precision and adaptability to changing environmental conditions. Factors such as soil nutrient variation, unpredictable weather patterns, and market fluctuations are not systematically analyzed, which can lead to reduced productivity and inefficient resource utilization.

2.1 Experience-Based Decision Making

In traditional farming, farmers depend heavily on personal experience and local knowledge to choose crops and plan cultivation activities. Although this approach is valuable, it does not always consider real-time variations in soil quality or climate conditions. As a result, farmers may select crops that are not optimally suited to the current environment, leading to lower yields and economic loss.

2.2 Lack of Data-Driven Insights

Traditional methods generally do not involve the use of scientific data such as soil nutrient analysis, weather forecasting, or market trends. Without proper data analysis, it becomes difficult to predict crop performance or respond to environmental changes. This limitation reduces the efficiency of farming practices and prevents farmers from making informed decisions that could improve productivity and profitability.

Ultimately, while traditional farming relies on invaluable ancestral knowledge, its lack of precision makes it increasingly vulnerable to modern environmental and economic shifts. The absence of real-time data and scientific validation often results in a "trial-and-error" approach to resource management. Consequently, there is a critical need for a more systematic, data-driven framework that can provide actionable insights. Such a transition is essential to optimize crop yields and ensure long-term agricultural sustainability in an unpredictable climate.

2.3 Motivation

The limitations of traditional farming methods highlight the need for a more advanced and data-driven approach. There is a strong demand for systems that can analyze soil, weather, and market conditions to provide accurate and timely recommendations. Motivated by this need, the proposed system aims to integrate modern technologies such as machine learning and data analytics to assist farmers in making better decisions. By providing crop recommendations and stage-wise advisory, the system helps improve productivity, reduce risks, and promote sustainable farming practices.

Chapter 3

Problem Statement and Objectives

Farmers often face challenges in selecting suitable crops due to variations in soil conditions, changing weather patterns, and lack of access to reliable information. Traditional farming decisions are usually based on experience, which may not always result in optimal yield or profit. There is a need for a system that can analyze multiple factors such as soil nutrients, environmental conditions, and market trends to support better decision-making. The objective of this project is to develop a smart crop recommendation system that suggests suitable crops based on soil and weather parameters, and provides stage-wise advisory to guide users throughout the crop lifecycle. The system also aims to incorporate weather-based insights and market information to help users make informed decisions regarding crop management and selling. Additionally, the project focuses on creating a user-friendly interface to ensure accessibility and ease of use

3.1 Problem Statement

Farmers often face difficulties in selecting suitable crops due to variations in soil conditions, changing weather patterns, and lack of access to reliable information. Traditional farming decisions are usually based on experience or general practices, which may not always lead to optimal yield or profit. Additionally, farmers lack proper

guidance throughout the crop lifecycle, including stages like planting, growth, and harvesting.

There is a need for an intelligent system that can analyze multiple factors such as soil nutrients, climate conditions, and market trends to provide accurate crop recommendations and stage-wise guidance. Such a system should be simple, accessible, and capable of supporting farmers in making informed decisions to improve productivity and sustainability.

3.2 OBJECTIVES

1) To develop a system that recommends suitable crops based on soil nutrients (N, P, K), pH, and environmental conditions.
2) To provide stage-wise crop advisory to guide users throughout the crop lifecycle.
3) To incorporate weather-based insights for better planning and decision-making.
4) To provide market insights to help farmers make better decisions for selling their products.
5) To design a user-friendly interface for easy interaction and accessibility.

In conclusion, the proposed system addresses the challenges faced in traditional farming by providing a data-driven solution for crop selection and management. By analyzing soil conditions, weather factors, and market information, the system helps users make more informed and effective decisions. The integration of crop recommendation and stage-wise advisory ensures that users are guided throughout the entire crop lifecycle. Overall, the system demonstrates how technology can be utilized to improve productivity, efficiency, and decision-making in farming, making it more reliable and systematic.

Chapter 4

Design and Implementation

The project is designed as an Artificial Intelligence (AI)-driven Crop Lifecycle Decision Support System tailored specifically for regional farmers. Architecturally, it is structured as a centralized web application that provides a personalized user dashboard for tracking farming activities. The core design philosophy centers on utilizing user-provided environmental data—such as soil nutrients (Nitrogen, Phosphorus, Potassium, pH) and local weather conditions—to drive intelligent agricultural decisions. By integrating customized predictive models and location-based market features, the system aims to recommend the most suitable crops, estimate expected yields and potential profits, and seamlessly connect farmers with optimal nearby seed shops based on proximity and competitive pricing.

On the implementation side, the system is built using the Python-based Django web framework for robust backend routing, secure user management, and HTML template rendering. Machine learning capabilities are woven directly into the application flow through dedicated utility scripts that load pre-trained predictive models to instantly process soil and weather inputs. State and relational data, including farmer profiles, crop tracking statistics, and geographical seed shop inventories, are managed via a lightweight SQLite database. The setup relies on automated data generation and seeding scripts to populate the system with localized agricultural and market data,

ensuring a realistic and ready-to-use environment for the end user.

4.1 DESIGN

The system is designed as a multi-tiered web application where the User Interface Module, Backend Processing Module, and Database Module operate in a synchronized closed-loop architecture. The process begins with the User/Farmer Input, which is captured via a secure Register/Login gateway that directs the user to a personalized Dashboard. This dashboard serves as the central navigation hub for four specialized sub-modules: the Crop Recommendation Module, the Market Advisory Module, the Weather Module, and the Crop Planner Module.

When a user submits environmental parameters—such as soil nitrogen levels, pH, or district location—the User Interface Module facilitates a bidirectional data flow with the Backend Processing Module. This backend layer acts as the system's backbone, where Python-based logic coordinates with an Admin Module for system oversight and manages high-frequency queries to the Database Module. Specifically, the Weather Module and Crop Recommendation Module feed data into the Crop Planner, ensuring that yield estimations are grounded in real-time climate and soil analysis. These insights are then processed by the backend and rendered into dynamic HTML templates, delivering actionable intelligence back to the farmer's dashboard in a clear, interactive format that simplifies complex agricultural data into practical decision-making steps.

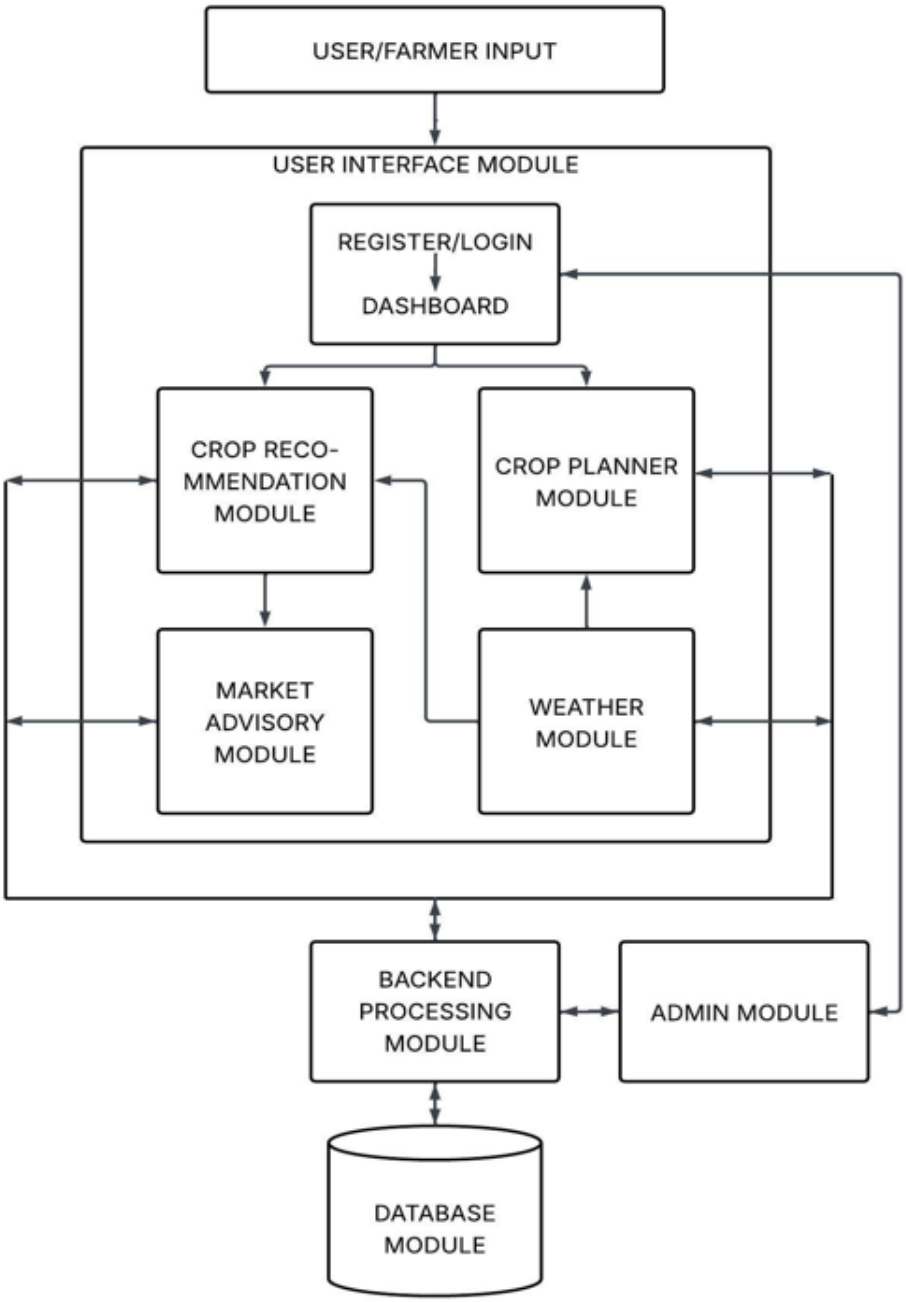

Figure 4.1: System Architecture Diagram

4.1.1 Level 0 DFD: Context Diagram

The Level 0 DFD represents the high-level scope of the Smart Crop Advisory System, illustrating the primary data exchange between the system and its users. The user inputs soil characteristics and crop preferences, which the system processes by interacting with external datasets. The cycle concludes when the system delivers specific crop-related recommendations back to the user, providing a clear boundary for the system's operations.

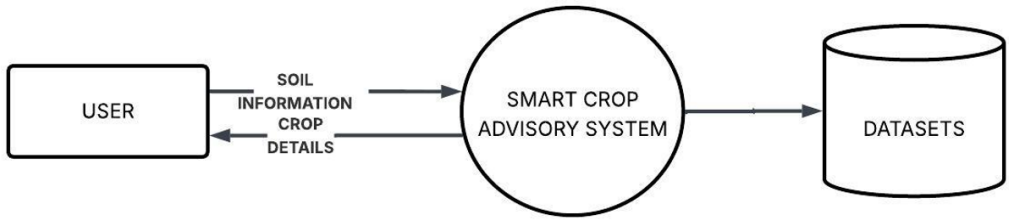

Figure 4.2: Level 0 Data Flow Diagram

4.1.2 Level 1 DFD: System Overview

The Level 1 DFD partitions the system into four main functional areas: Authentication, Crop Tracker, Crop Recommendation, and Market Recommendation. Upon successful identity validation through the user database, the system allows the user to either monitor existing crops or receive new suggestions based on environmental factors. This diagram introduces the integration of weather forecast data, which serves as a critical external variable for both the tracking and recommendation processes.

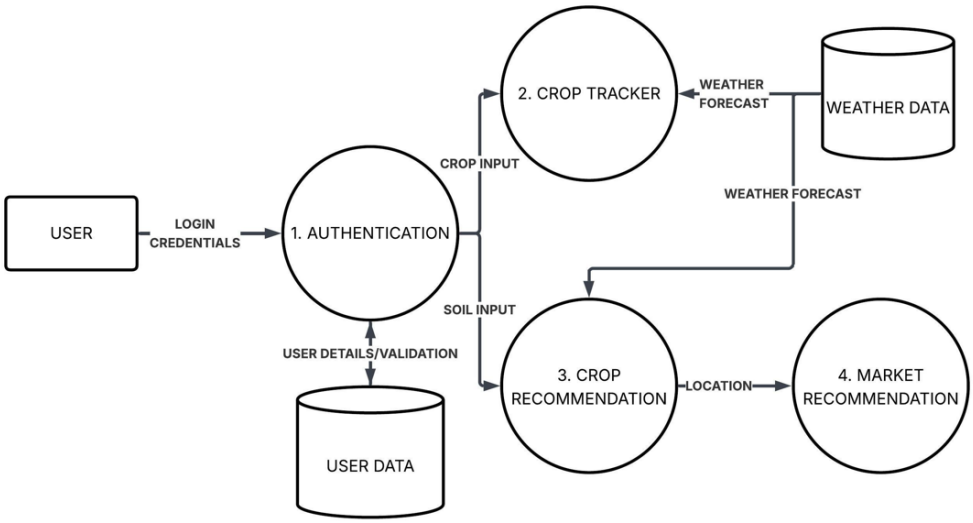

Figure 4.3: Level 1 Data Flow Diagram

4.1.3 Level 2 DFD (Process 2): Crop Tracker

This Level 2 DFD describes the internal logic of the Crop Tracker module, focusing on the monitoring of agricultural progress. The "Analyse Crop" process receives soil profiles and matches them against established crop growth stages and real-time weather data. The matching algorithm evaluates these inputs to generate a status report, which is then used to maintain an updated record of the crop's development and health.

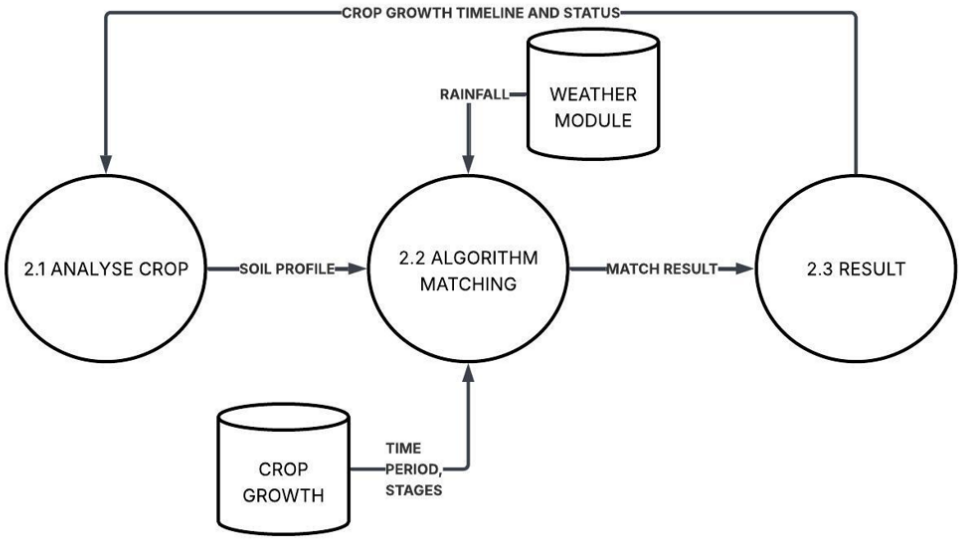

Figure 4.4: Level 2 Data Flow Diagram for Crop Tracker Process

4.1.4 Level 2 DFD (Process 3): Crop Recommendation

The Level 2 DFD for Crop Recommendation outlines the filtering process used to identify optimal crops for a specific location. The system first evaluates soil properties and temperature data to determine biological compatibility. These results are then refined by a secondary filter that incorporates seed rates, historical market trends, and yield projections to ensure that the final recommendations are both feasible and economically advantageous for the user.

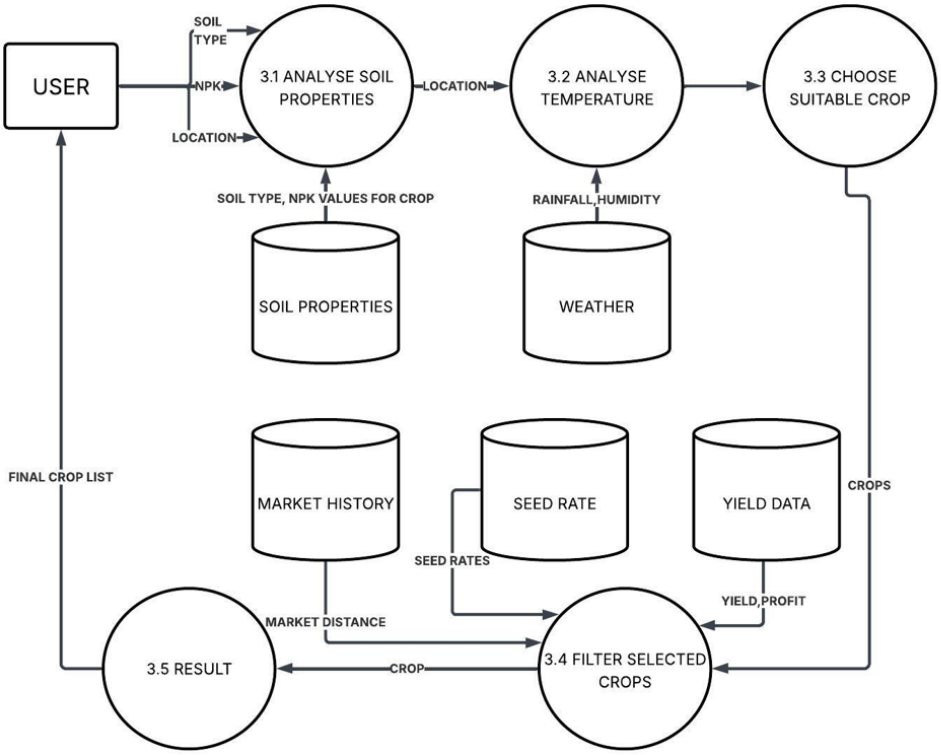

Figure 4.5: Level 2 Data Flow Diagram for Crop Recommendation Process

4.1.5 Level 2 DFD (Process 4): Market Recommendation

The final Level 2 DFD details the Market Recommendation module, designed to assist users in identifying the most suitable sales outlets. By analyzing the crop type and current seed data, the system identifies potential markets through historical and real-time data stores. These options are then filtered based on transportation distances and production costs to provide the user with a prioritized list of markets that maximize potential profit.

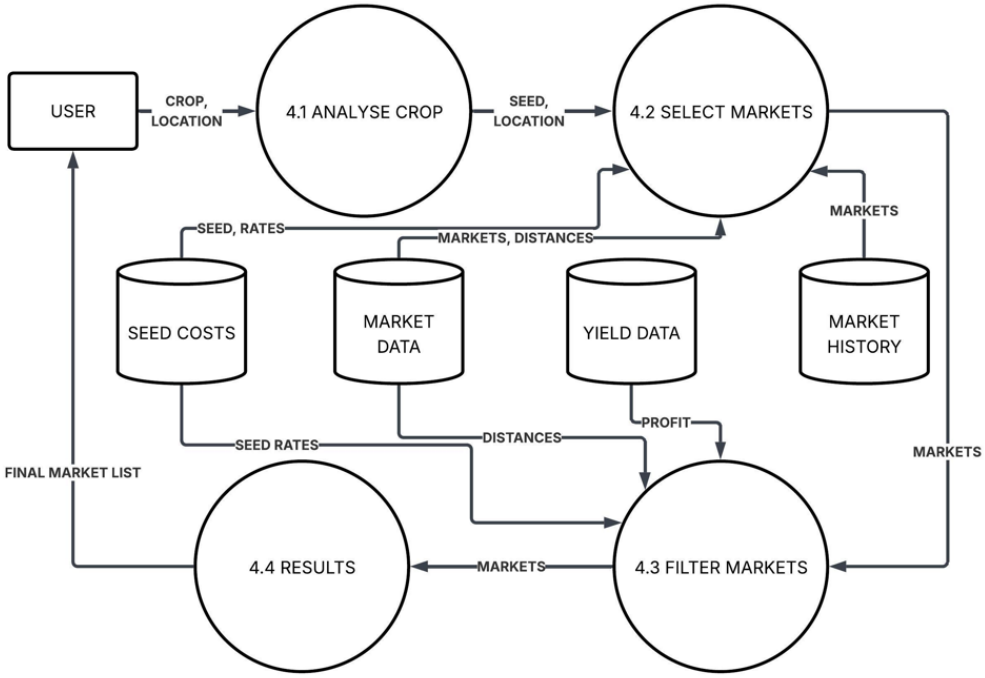

Figure 4.6: Level 2 Data Flow Diagram for Market Recommendation Process

4.2 IMPLEMENTATION

The system is implemented using Python and Django for the backend, with HTML, CSS, and JavaScript for the frontend interface. Machine learning models are developed using libraries such as Scikit-learn, where a Random Forest algorithm is trained on agricultural datasets to perform crop prediction. The trained models are saved as .pkl files and loaded during runtime for making predictions. Datasets are stored in CSV format and are used for various functionalities such as crop recommendation, yield estimation, and advisory generation. The advisory system is implemented using a dataset-driven approach, where crop stages and corresponding descriptions are stored with defined day ranges. The system dynamically calculates the number of days since planting and retrieves the appropriate stage and advice. Additional features such as weather-based alerts, market insights, and crop timelines are implemented

using modular functions to ensure scalability and maintainability. Overall, the implementation focuses on integrating machine learning, data processing, and user interface design to create an efficient and practical solution for smart farming.

In summary, the design and implementation of this system create a cohesive bridge between complex machine learning algorithms and practical agricultural management. By leveraging the Django framework and Scikit-learn models, the architecture ensures that data-driven insights are both accurate and accessible to the end user. This technical foundation provides a scalable and robust environment capable of evolving with changing environmental data. Ultimately, the integrated setup transforms raw soil and weather inputs into a functional tool for enhancing farm productivity and decision-making.

Chapter 5

Results and Discussion

The developed system successfully provides crop recommendations based on soil nutrients, pH, and environmental conditions with satisfactory accuracy. The machine learning model predicts suitable crops for various input combinations, and the results are consistent with expected agricultural practices. The advisory module effectively delivers stage-wise guidance, helping users understand the crop lifecycle and required actions at each stage. Additional features such as weather-based insights and market information further enhance the usefulness of the system. The results demonstrate that integrating machine learning with agricultural data can significantly improve decision-making compared to traditional methods. However, the performance of the system depends on the quality and diversity of the dataset used. While the current implementation produces reliable outputs, it can be further improved by incorporating real-time data, expanding the dataset, and using more advanced models. Overall, the project highlights the potential of data-driven approaches in making farming more efficient, scalable, and informed.

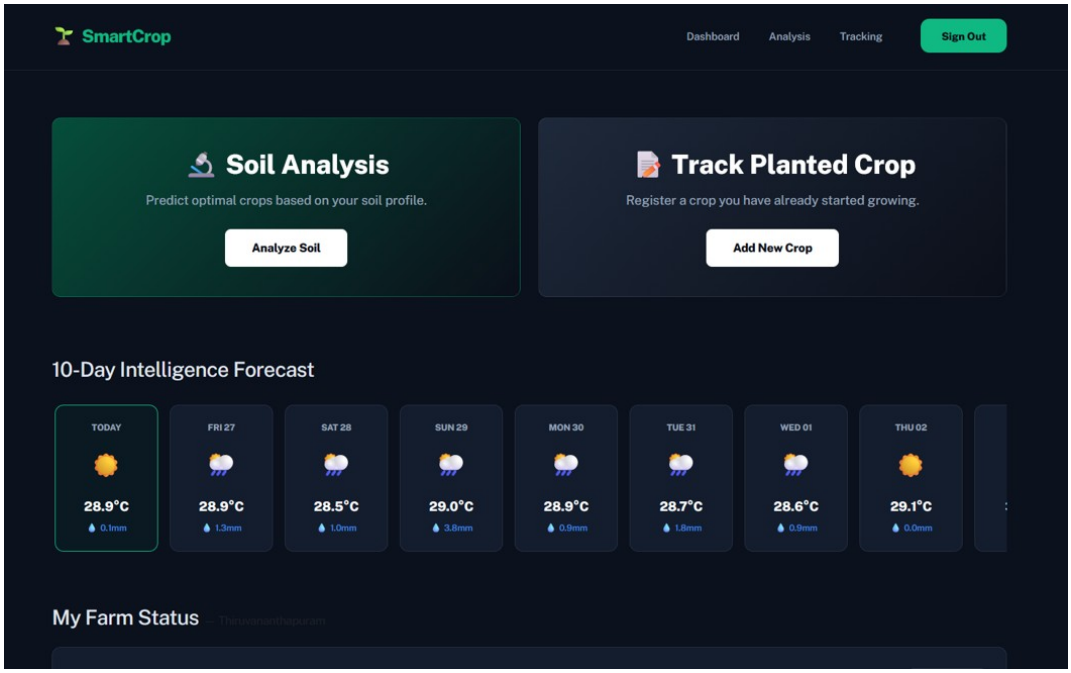

Figure 5.1: Dashboard fig 1

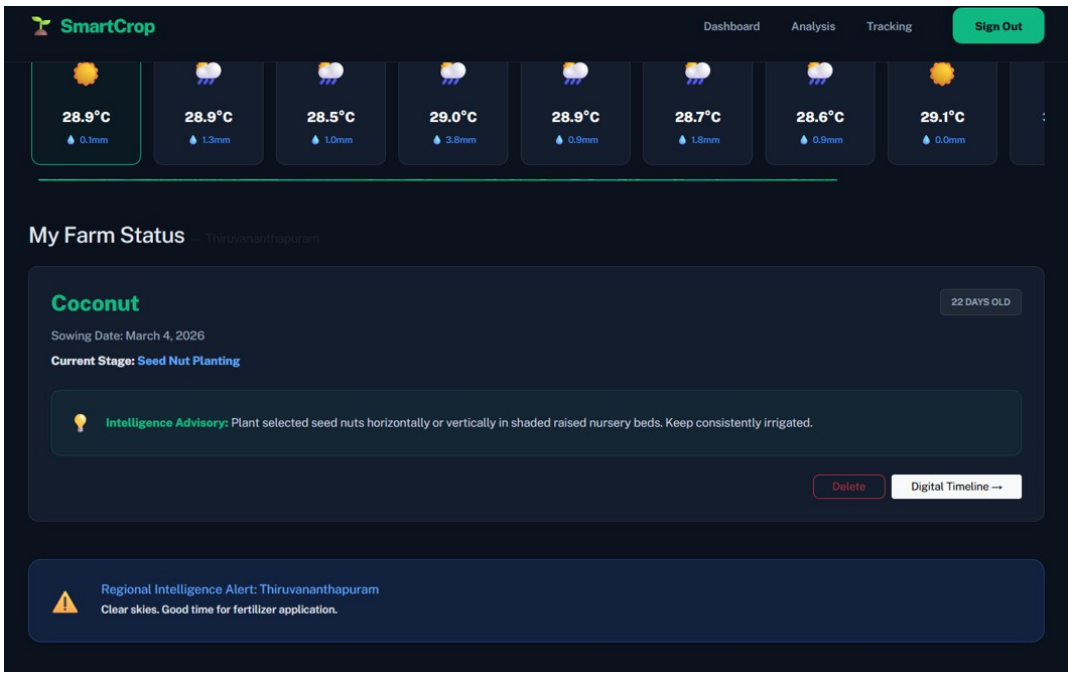

Figure 5.2: Dashboard fig 2

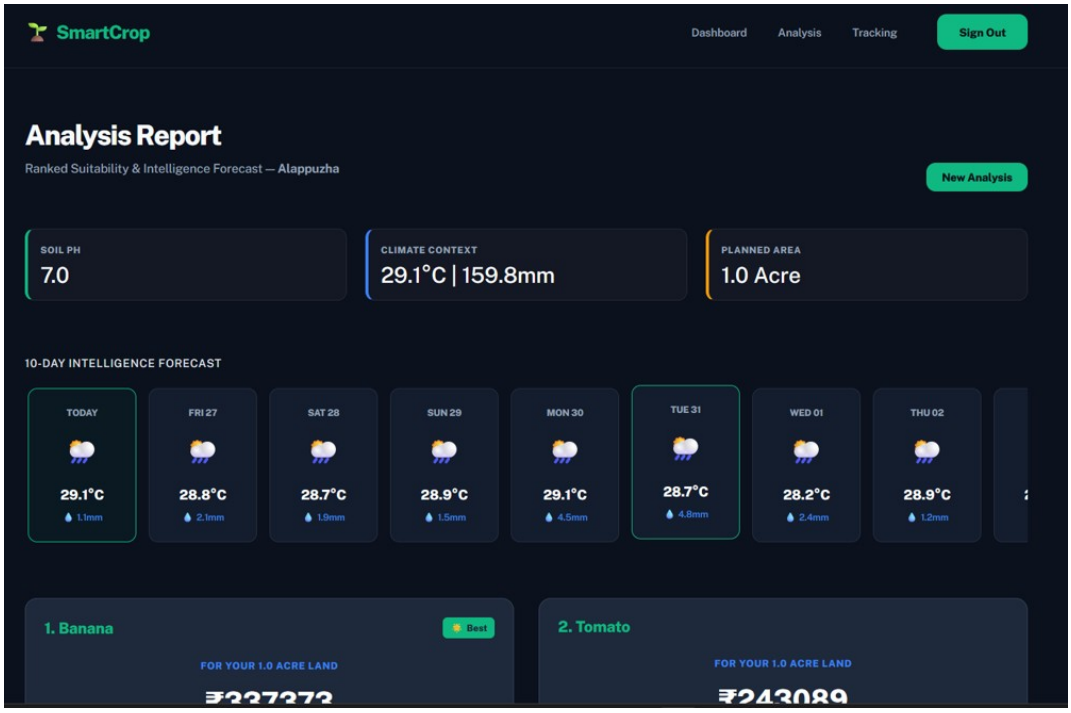

Figure 5.3: Crop Recommendation page fig 1

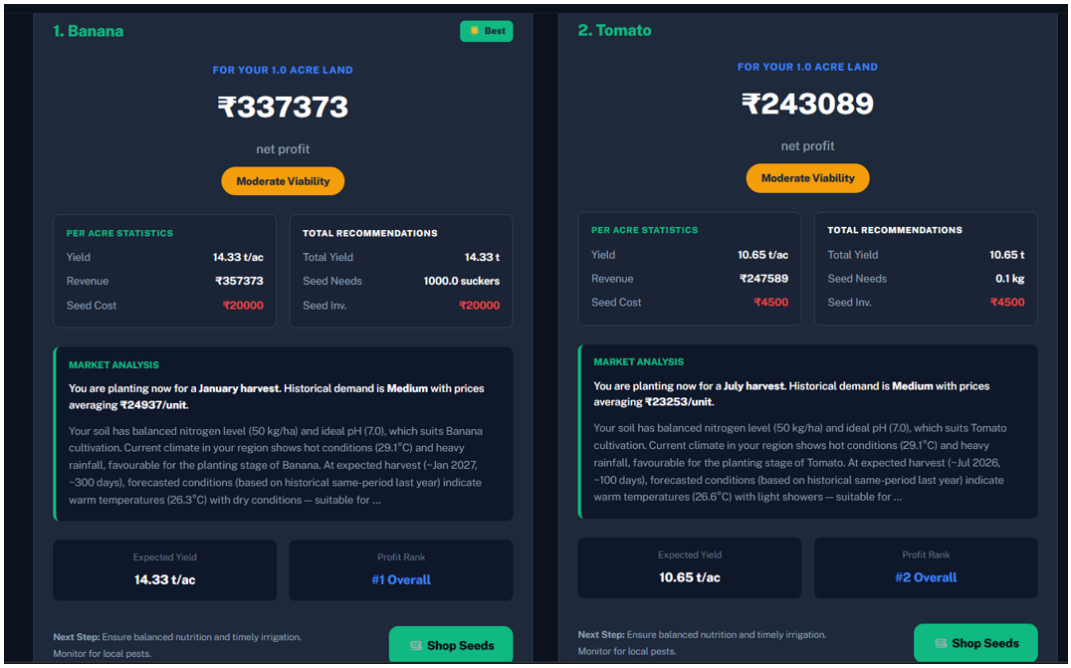

Figure 5.4: Crop Recommendation page fig 2

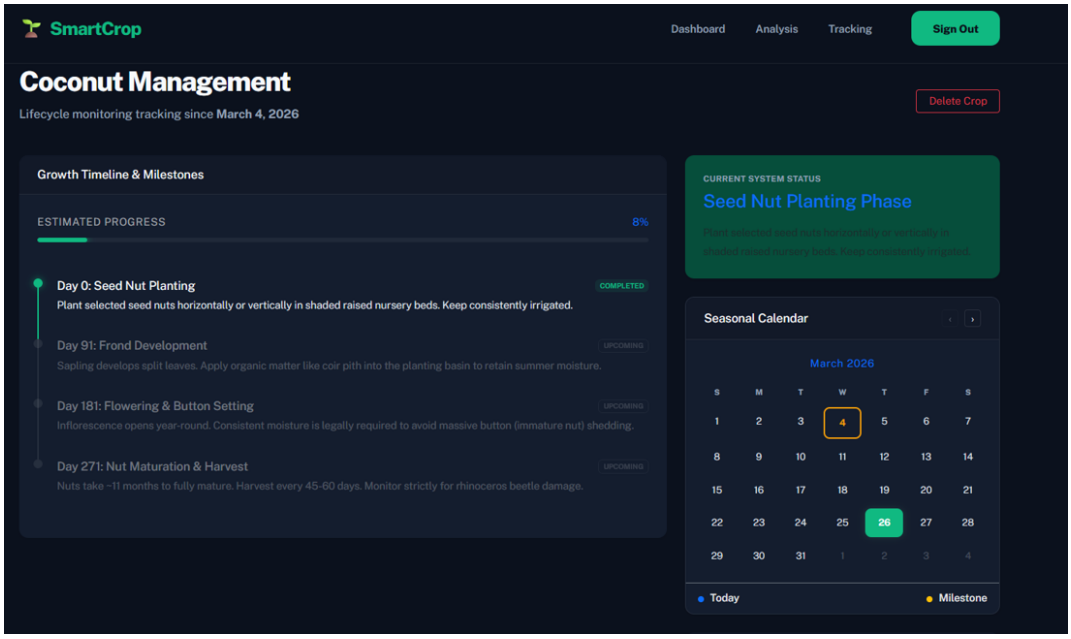

Figure 5.5: Crop Tracker page

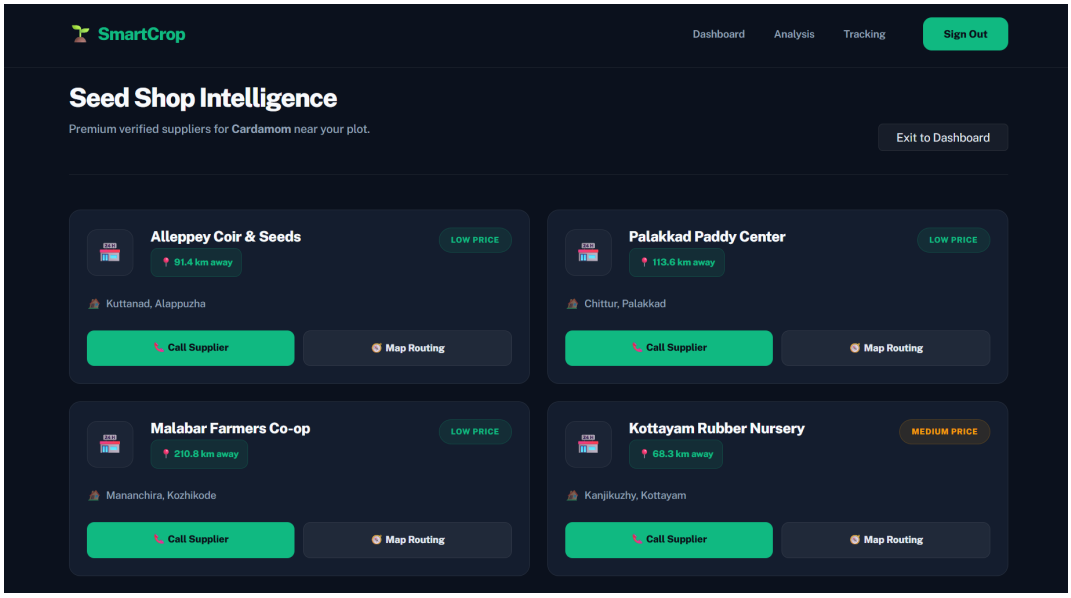

Figure 5.6: Market Recommendation page

Chapter 6

Conclusion and Future Scope

The development of the Smart Crop Recommendation System marks a significant transition from traditional, intuition-based farming to a high-precision, data-driven agricultural model. By leveraging machine learning—specifically the Random Forest algorithm—the system successfully transforms complex environmental variables, such as soil nutrients (N, P, K), pH levels, and climatic conditions, into actionable intelligence. Beyond mere selection, the integration of stage-wise advisory features and yield prediction ensures that farmers are supported throughout the entire crop lifecycle, effectively addressing the challenges posed by unpredictable weather and market fluctuations.

Overall, this project demonstrates how AI-driven tools can bridge the gap between scientific analysis and field-level application, making farming more efficient, informed, and resilient.Looking forward, the system is designed with scalability in mind to adapt to the evolving needs of modern agriculture. Future iterations will focus on developing a dynamic crop planner that recalibrates recommendations in real-time based on shifting weather patterns and specific growth stages. The user experience will be further enhanced through a map-based interface for intuitive district selection and the integration of real-time location services to provide localized market insights. By expanding the dataset to include a wider variety of regional crops

and incorporating advanced price prediction models, the system aims to not only optimize productivity but also maximize the economic profitability for farmers in an increasingly competitive global market.

References

[1] AI-Driven Platform for Crop Optimization, Weather Forecasting and Agricultural Innovation, *B. Sribharathi; V. Cibirajan; S. Harikishore; S. Sakthivel, ICRDICCT, 2025.*

[2] Perfect Crop Recommendation System Using ML, *Avinash Helkar; Ashish Manure; Sanket Mhetre; Uday Patil; Sahebgouda Byle; M. D. Katkar, IRJMETS, November 2025.*

[3] Smart Agriculture Marketplace For Enhanced Farming And E-Commerce Connectivity, *Shankar Gadhve; Ajinkya Jawade; Samiksha Vairagade; Pranjal Kale; Uday Rudrakar, IJCRT, Volume 13, Issue 4, April 2025.*

[4] Smart Crop Advisory Systems Using Artificial Intelligence and Machine Learning, *Sneh Lata Singh; Amit Yadav; Shanu Ahmed; Abhay Maurya; Varun Rana; Samir Ahmad, Indian Journal of Computer Science and Technology, September–December 2025.*

[5] Determinants of Small Farmers' Access to Agricultural Markets: Evidence from South India, *R. L. Manogna, Discover Sustainability, 2025.*

[6] High-Precision Crop Recommendation System with Stacking Ensemble Classifiers for Optimizing Agricultural Productivity, *Rania A. Ahmed; Walid El-Shafai; Zeinab A. Ahmed; El-Sayed M. El-Rabaie; Fathi E. Abd El-Samie et al., Scientific Reports, 2025.*

[7] AI-based Smart Crop Recommendation System for Sustainable Agricultural Production: A Data-driven Approach to Minimize Resource Use and Maximize Yield, *Aarthi S.; Manimegalai S.; Sakthivel R., Madras Agricultural Journal, 2025.*

[8] Crop Recommendation With Machine Learning: Leveraging Environmental And Economic Factors For Optimal Crop Selection, *Steven Sam; Silima Marshal D'Abreo, arXiv, May 2025.*

[9] Scientific Planning of Dynamic Crops in Complex Agricultural Landscapes Based on Adaptive Optimization Hybrid SA-GA Method, *Changlong Li; Zengye Su; Yudan Nie; Zhiyi Ye; Jinyi Li; Jing Wang; Zicong Yang; Xuxin Li; Weijian Zeng; Yanjian Chen, Scientific Reports, 2025.*

[10] Smart Farming System – An Efficient Technique by Predicting Agriculture Yields Based on Machine Learning, *D. Harinath; Archana Patil; Madhu Bandi; Divya Spandana, Technische Sicherheit, 2024.*

[11] Predicting Crop Disease Outbreaks Using Weather and Soil Data: A Machine Learning Approach to Risk-Based Crop Protection, *O. G. Emi-Johnson; O. O. Fasanya; T. K. Amusa, IJETRM, Volume 08, Issue 03, March 2024.*

[12] Enhancing Agricultural Productivity: A Machine Learning Approach to Crop Recommendations, *Farida Siddiqi Prity; Md. Mehadi Hasan; Shakhawat Hossain Saif; Md. Maruf Hossain; Sazzad Hossain Bhuiyan; Md. Ariful Islam Lavlu, Human-Centric Intelligent Systems, 2024.*

[13] AI-based Market Intelligence Systems for Farmer Collectives: A Case Study from India, *Ronak Ladhar; Sourav Sharma; Nishant Gupta; Rishi Shah; Aditya Bommakanti; Arjav Jain; Bhuvan Chand Katakam; Dipanjan Chakraborty; Hari Prasad Piridi; Kaushalendra Yadav; Kumra Vittalrao; Matiur Rahman; C.*

Sankaraiah; Rohan Katepallewar; Abdul Azeem; Srikant Tangirala, ACM Journal on Computing and Sustainable Societies, Volume 1, Issue 1, September 2023.

[14] Government Agriculture Schemes in India: A Review, *Gagan Tripathi; Arpit Dhodia; Anmol Giri; Veena Rathore; Aman Verma; Anoop Shukla; Lalit Kumar Verma, Asian Journal of Agricultural Extension, Economics & Sociology, 2023.*

[15] Agriculture Marketing in India: Perspectives on Reforms and Doubling Farmers' Income, *Debesh Roy, Journal of Marketing Development and Competitiveness, Volume 16(3), 2022.*

[16] Optimizing Crop Planting Schedule Considering Planting Window and Storage Capacity, *Saiara Samira Sajid; Guiping Hu, Frontiers in Plant Science, March 2022.*

[17] Price Forecasting and Market Intelligence for Agricultural Commodities Using Machine Learning, *R. L. Manogna, Agricultural Economics Research Review, 2021.*

[18] Small and Marginal Farmers of Indian Agriculture: Prospects and Extension Strategies, *Sunil Kumar; D. U. M. Rao; Pratibha Thombare; Pandurang Kale, Indian Research Journal of Extension Education, Volume 20(1), January 2020.*

[19] Crop Disease Detection using Machine Learning: Indian Agriculture, *Anuradha Badage, IRJET, Volume 05, Issue 09, September 2018.*

[20] Crop Recommendation System to Maximize Crop Yield using Machine Learning Technique, *Rohit Kumar Rajak; Ankit Pawar; Mitalee Pendke; Pooja Shinde; Suresh Rathod; Avinash Devare, IRJET, Volume 04, Issue 12, December 2017.*